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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "05-07-2026_13:36:13",
  "variant" : 2,
  "subjectName" : "mathematics",
  "categories" : [ "Mathematical Logic", "Stereometry", "Combinatorics" ],
  "questions" : [ {
    "question" : "What is the negation of the statement 'All primes are odd'?",
    "answers" : [ "Some primes are odd", "All primes are even", "No primes are odd", "There exists at least one prime that is not odd" ],
    "correctIndex" : 3
  }, {
    "question" : "If P is true and Q is false, what is the truth value of the implication  $P \rightarrow Q$ ?",
    "answers" : [ "False", "Contingent", "Undefined", "True" ],
    "correctIndex" : 0
  }, {
    "question" : "What is a logical statement called if it evaluates to true under all possible truth value assignments?",
    "answers" : [ "Contingency", "Fallacy", "Tautology", "Contradiction" ],
    "correctIndex" : 2
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    "question" : "What logic law simplifies 'not (A and B)' into '(not A) or (not B)'?",
    "answers" : [ "Associative Law", "De Morgan's Law", "Commutative Law", "Distributive Law" ],
    "correctIndex" : 1
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    "question" : "How many vertices does a standard cube possess?",
    "answers" : [ "16", "6", "8", "12" ],
    "correctIndex" : 2
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    "question" : "What is the volume formula for a perfect sphere with radius r?",
    "answers" : [ " $4 * \pi * r^2$ ", " $(4/3) * \pi * r^3$ ", " $2 * \pi * r$ ", " $\pi * r^2$ " ],
    "correctIndex" : 1
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    "question" : "In how many ways can 4 people be seated in a straight line?",
    "answers" : [ "24", "16", "6", "12" ],
    "correctIndex" : 0
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    "question" : "How many distinct subsets can be formed from a set containing 5 elements?",
    "answers" : [ "25", "64", "16", "32" ],
    "correctIndex" : 3
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    "question" : "What is the value of the combinatorial coefficient  $C(5, 2)$ ?",

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"answers" : [ "20", "15", "5", "10" ],
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"correctIndex" : 3
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}, {
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"question" : "How many unique permutations can be made from the letters in the word 'APP'?",
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"answers" : [ "3", "6", "1", "9" ],
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"correctIndex" : 0
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}]
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}
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